Rebuilding the Blocks:
Firm Heterogeneity and Urban Retail After Remote Work

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Job Market Paper

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Introduction

- Urban locations differ in the number *and quality* of restaurants and shops.
  - Agglomeration literature: why and where firms cluster.
  - Heterogeneous firm quality is unexplored.
- Concern: work-from-home (WFH) leaves the city center deserted.
  - Foot traffic from office workers is local demand for retail.
  - San Francisco: ∼35% primarily WFH in 2021, about twice the U.S. rate.
  - *If* firms are leaving (lost demand vs. lower rents), and *which* firms are leaving (marginal firms).

This paper

- How do spatial demand, competition, and spillovers shape the distribution of heterogeneous retail firms within and across locations?
  - Quantitative spatial equilibrium framework with heterogeneous firms.
- What happens to the city center under work-from-home?
  - Counterfactuals: shift x% of persons to work from home.
- Preview of results
  - City center: job-dense core gains entrants (+3.6% at 50% WFH).
  - Welfare: consumer surplus rises 2.7% to 8.0% (purely from better access to retail).

Literature

- Amenities and urban spatial equilibrium
  Ahlfeldt et al. '15; Oh Seo '23; Almagro Domínguez-Iino '25; Miyauchi Nakajima Redding '25
  → Heterogeneous firm quality within locations, not just firm counts.

- Firm location choice and spatial competition
  Seim '06; Zhu Singh '09; Holmes '11
  → Endogenous location quality, with multiple sectors and trip-chaining.

- GPS data in urban economics
  Athey et al. '18; Cao et al. '24; Cook '24; Couture Dingel et al. '25; Nguyen Tran '25
  → Identify supply from within-location demand estimates.

- Remote work and cities
  Liu Su '21; Ramani Bloom '21; Rosenthal Strange Urrego '22; Davis Ghent Gregory '24; Qian Su '25
  → Quality-dependent response of retail firms to remote work.

Data

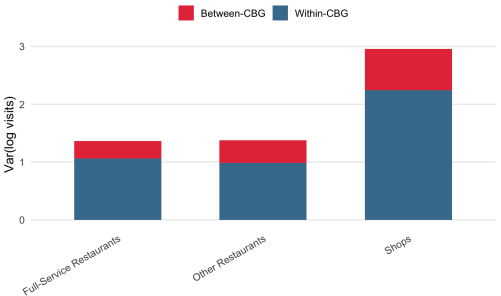
- San Francisco, June 2019 (pre-pandemic)
- Veraset mobile-device GPS data
  - Visits: device, home block group, time, dwell time, establishment.
  - Trips: consecutive visits within a 3-hour window.
  - Work location inferred from long weekday daytime stays.
- Sales tax by tax block group (SF Controller)
  - Validates visits: 1% more visits $\rightarrow$ 0.26% more sales tax.
- Business registry (SF Treasurer & Tax Collector)
- Sample
  - Location = Census block group (CBG), about 500.
  - About 4,000 restaurants and 3,500 shops.

Summary statistics

	Mean	P25	Median	P75
Block group level				
Establishments	15.3	2.5	8	19
Restaurants	8.5	1	4	11
Shops	6.8	1	3	7.5
Total visits	3,367	492	1,271	3,432
Establishment level				
Restaurant visits	262.7	81	164	289
Shop visits	167.2	17	68	178
Restaurant dwell time (min)	136.2	73	109	164

- Firm counts per block group range from 1 to over 300.
- Restaurants account for about 60% of visits.
- About 25% of block groups have at least one chain restaurant; 36% have at least one chain shop.

Descriptive evidence 1: variance decomposition





- Most variation in firm visits is within block groups (left).
  - Full-service restaurants 78%, other restaurants 72%, shops 76%.
- Within share rises with location popularity (right).
  - Better locations host a wider range of firms, not necessarily uniformly better firms.

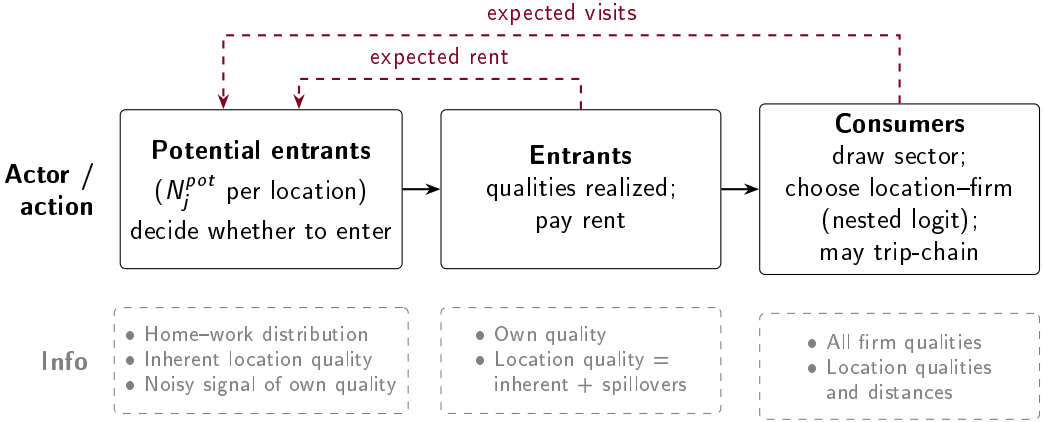
Descriptive evidence 2: trip-chaining

Transition probabilities between consecutive stops in a trip

First stop \ Next stop	Full-service	Other restaurants	Shops
Full-service restaurants	0.507	0.204	0.289
Other restaurants	0.402	0.300	0.298
Shops	0.377	0.189	0.435

- Consumers make 1.58 visits per trip on average.
- 32% of next stops are in the same block group.

Model overview



- City of locations with residents and workers (home–work flows taken as given).
- Two sectors: restaurants and shops.

Demand model

- Consumer living in h and working in w draws a sector s, then chooses a location j, then a firm k.
- Firm choice within a location (δ_{sjk} = firm quality, ρ = nesting parameter):

$$s_{k|sj} = \frac{\exp(\delta_{sjk}/\rho)}{\sum_{k'} \exp(\delta_{sjk'}/\rho)}$$

- Location choice:

$$\delta_{sjhw} = \gamma_{shw} + \beta^{home}\,\text{Dist}_{hj} + \beta^{work}\,\text{Dist}_{wj} + \beta^{TC}\,\text{TripChain}_{sj} + \text{Estab}_{sj} + \xi_{sj}$$

  - Estab_{sj}: inclusive value of firms in j.
  - ξ_{sj}: location quality.
  - TripChain_{sj}: option value of a second stop, given first-stop sector s and location j.

Supply model

- Location quality = fundamentals (inherent) + economies-of-scale spillovers:

$$\xi_j = \tilde{\delta}_j^{loc} + \alpha N_j$$

- Profit of firm k in location j:

$$\pi_{jk} = \beta \log \text{visits}_{jk} - \gamma_0 \underbrace{- \gamma_1 \log w_j - \gamma_2 \log r_j - \gamma_3 \log N_j}_{\sim \text{ rent}}$$

- Heterogeneous quality and incomplete information
  - True quality $\delta_{jk} \sim N(\mu, \sigma_\delta^2)$.
  - Noisy signal $\tilde{\delta}_{jk} = \delta_{jk} + \nu_{jk}$, $\nu_{jk} \sim N(0, \sigma_\nu^2)$.
  - Potential entrants form expectations on visits_{jk} and N_j.

Equilibrium

- Entry follows a threshold rule: enter if $\tilde{\delta}_{jk} \geq \tau_j^*$.
- Equilibrium: thresholds such that the marginal entrant earns zero expected profit in every location.
  - Existence by Brouwer's fixed-point theorem.
  - Multiple equilibria; select by starting from observed entry.
- Why heterogeneity matters
  - Marginal firms are the lowest quality and least visited *in expectation*.
  - Net entry is ambiguous: demand and rent move in opposite directions, interacting with spillovers.

Identification and estimation

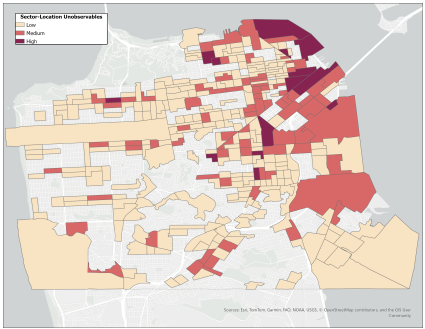
- Demand
  - Gravity equations for location choices and trip sequences.
  - Firm quality from market shares relative to a chain reference firm.
- Supply, step 1: maximum likelihood
  - Fix N_j^{pot}.
  - Entry counts and entrants' quality distribution give thresholds and $(\mu, \sigma_\delta, \sigma_\nu)$.
- Supply, step 2: simulated method of moments
  - Zero-profit condition at each location's threshold gives (α, β, γ).
- Identification idea
  - Fundamentals and spillovers both raise location quality and firm counts.
  - But fundamentals also change the quality distribution of entrants.

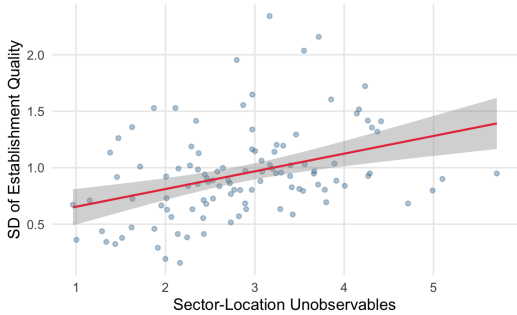
Estimation results: demand

	Destination choice		Trip-chaining choice	
	Restaurants	Shops	Restaurants	Shops
Same home CBG	0.315	0.251		
Distance from home	−0.073	−0.046		
Same work CBG	1.340	1.085		
Distance from work	−0.049	−0.037		
Same previous CBG			2.085	1.748
Distance from previous			−0.447	−0.328

- All coefficients significant at 1%; quadratic distance terms and location FEs included.
- Strong pull of workplace and previous stop.
- Trip-chaining coefficient $\hat{\beta}^{TC} = 0.06$, smaller than Oh & Seo (2023).

Estimation results: location and firm quality





- Left: location quality $\hat{\xi}$ for restaurants, highest in the northeast commercial core.
- Right: within-location dispersion of firm quality rises with location quality.
- Controlling for dispersion lowers the firm-count effect on $\hat{\xi}$ by 19% (restaurants), 10% (shops).

Estimation results: supply

Parameter	Estimate	SE
Mean quality μ	-1.84	
SD of quality σ_δ	1.67	
SD of signal noise σ_ν	1.74	
Agglomeration α	0.087	(0.006)
Revenue elasticity β	1.001	(0.019)
Worker rent γ_1	0.000	(0.031)
Resident rent γ_2	0.001	(0.033)
Fixed cost γ_0	-0.565	(0.213)

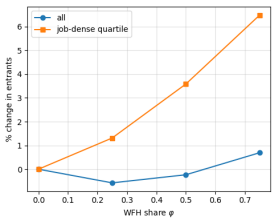
- Restaurants, potential entrants $= 2\times$ max in neighborhood.
- Entrants are a selected right tail; $\hat{\alpha}$ close to literature (mean 0.096).
- Profit parameters sensitive to the entrant pool; counterfactuals use $\alpha = 0$.

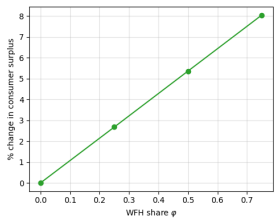
Counterfactual: WFH adoption levels

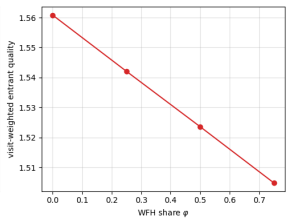
- Shift a share ϕ of persons to work from home.
  - $\phi = 25\%, 50\%, 75\%$
  - Residents unchanged.
- Two channels
  - Demand: trips start from homes instead of offices.
  - Cost: rents fall where workers leave.
- Re-solve entry thresholds in every location.
- Held fixed: quality distribution, profit parameters, fundamentals.

Counterfactual: aggregate effects

WFH share	Entrants (%)	Visits (%)	Consumer surplus (%)	Entrant quality
0% (baseline)	0.00	0.00	0.00	1.561
25%	−0.58	+0.38	+2.68	1.542
50%	−0.23	+0.76	+5.36	1.524
75%	+0.69	+1.15	+8.05	1.505







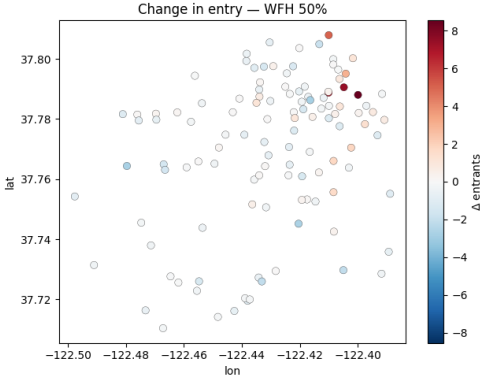
- Citywide firm count roughly unchanged; consumer surplus rises with WFH.

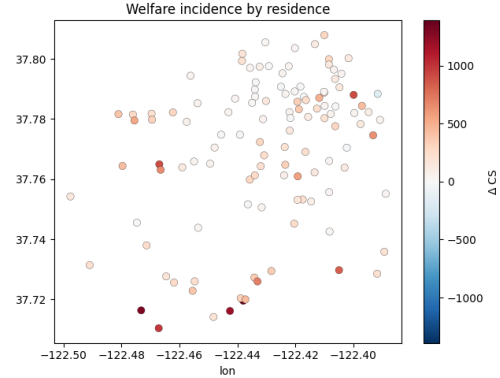
Counterfactual: the city center

Entrants	Baseline	50% WFH	Change
Job-dense core	1000.6	1036.5	+3.6%
Residential periphery	1036.8	996.2	−3.9%
City			−0.2%

- Why the core gains firms
  - Fewer workers lower demand, but rents fall too.
  - High location quality still attracts non-local visitors.
- Why the periphery loses firms
  - More workers raise demand, but also rents.

Counterfactual: spatial incidence at 50% WFH





- Entry gains concentrate in the northeast core; surplus gains go to residents of the south and west.

Counterfactual: decomposition and welfare

Scenario	Entrants (%)	Consumer surplus (%)
WFH 50%	−0.23	+5.36
Demand channel only	0.00	+5.45
Cost channel only	−0.26	−0.04
Placebo: scramble home–work pairs	0.00	−5.39

- Surplus gains come through demand: shorter trips to retail.
- Entry changes come through rents.
- Welfare measure excludes commuting savings, residential sorting, landlord income.

Conclusion

- Firm quality, not just firm counts, matters for urban retail.
  - 75% of firm-level variation is within block groups.
  - Consumers chain trips, linking demand across firms and sectors.
- Spatial equilibrium model with heterogeneous firms.
  - Better locations support a wider range of firm quality.
- WFH does not hollow out downtown retail.
  - Core gains 3.6% of entrants.
  - Consumer surplus rises 2.7% to 8.0%.
- What's next
  - Dynamics: entry *and exit*, rent stickiness.
  - Policy: rezoning, rent relief, office-to-residential conversion.
  - Broader welfare: commuting, residential sorting, landlords.

Thank you